HUM1012 : Logic and Language Structure

(LT)

Slot : B21 (Mon 04.25 pm – 05.55 pm) + E14 (Tue 02.50 pm – 04.20 pm)
D21 (Tue 01.15 pm – 02.45 pm) B14 + (Wed 02.50 pm – 04.20 pm)

Venue : AB02 – 301 (B21+E14)
AB02 – 301 (B14+D21)

Prepared and Presented by

Dr. Velmani Ramasamy, M.E., Ph.D., SMIEEE., MACM.,

Associate Professor (Grade 2)

School of SCAI,

VIT Bhopal University,

Bhopal, Madhya Pradesh.

MODULE 1 INTRODUCTION TO NATURAL LANGUAGE PROCESSING 3

Human language, models, ambiguity, processing paradigms, phases in natural language processing, need for NLP and NLU, relevance of NLP and NLU, understanding of language (natural language, symbolic language, computer language), application of natural language in programming, expert system in AI, recommendation system in AI and cognitive system in AI.

COURSE OUTCOMES:

CO1: Understanding approaches to syntax and semantics in NLP.

Natural Language Processing (NLP)

Natural Language Processing (NLP)

- NLP is a branch of AI that focuses on enabling computers to understand, interpret, generate, and interact using human language in a meaningful way.
- NLP combines linguistics, computer science, and machine learning to process and analyze large amounts of natural language data.
- NLP is the technology that allows machines to read, understand, and respond to human language.

Goals of NLP:

- ✓ Understand human language (text or speech)
- ✓ Extract meaning and context
- ✓ Communicate in human-like ways
- ✓ Automate language-based tasks

Common NLP Tasks:

Task	Description
Tokenization	Splitting text into words or phrases
Part-of-Speech Tagging	Identifying grammar components (noun, verb, etc.)
Named Entity Recognition (NER)	Detecting names, locations, organizations, etc.
Sentiment Analysis	Determining emotion or opinion in text
Machine Translation	Translating between languages (e.g., English to Hindi)
Text Classification	Categorizing content (e.g., spam detection)
Speech-to-text / Text-to-speech	via combined models
Summarization	Condensing long texts
Question answering	Search or Chatbots
Speech Recognition	Converting spoken words to text

NLP Applications:

- ✓ Virtual assistants (e.g., **Alexa**, **Siri**, **ChatGPT**)
- ✓ Language translation (**Google Translate**)
- ✓ Chatbots and customer service
- ✓ Search engines (**Google**, **Bing, Opera, Mozilla**) understanding queries
- ✓ Email filtering (e.g., spam detection)
- ✓ Social media monitoring

NLP Tools & Libraries:

- ✓ **Python** libraries: NLTK (Natural Language Tool Kit), spaCy, TextBlob
- ✓ **Deep learning** models:

  BERT (Bidirectional Encoder Representation from Transformers),

  GPT (Generative Pretrained Transformer),

  T5 (Text-to-Text Transfer Transformer) (via Hugging FaceTransformers)
- ✓ **Platforms**: OpenAI, Google Cloud NLP, AWS Comprehend

1.1 Human Language

Human Language

- Human language is a complex system of **communication** that is unique to humans.
- It's composed of **symbols (words or signs)** and **rules (grammar)** that allow people to express and understand **thoughts, emotions, ideas, and information**.

Human Voice Frequency

- The **human voice frequency range** refers to the range of sound frequencies produced during speech.
- It's usually measured in **Hertz (Hz)** and varies by age, gender, and vocal type.

Human Voice Frequency Ranges

Voice Type	Frequency Range (Approx.)
Adult Male	85 Hz – 180 Hz
Adult Female	165 Hz – 255 Hz
Children	250 Hz – 400 Hz (and higher)

Full Speech Frequency Range

Purpose	Frequency Range
Phone quality speech	~300 Hz – 3.4 kHz
Studio-quality speech	~80 Hz – 14 kHz
Human hearing range	~20 Hz – 20 kHz

Applications of Voice Frequency Analysis

- ✓ Speech recognition
- ✓ Voice biometrics (e.g., speaker ID)
- ✓ Emotion detection
- ✓ Health diagnostics (e.g., Parkinson's, stress)
- ✓ Audio compression (e.g., MP3, VoIP)

Features of Human Language

(i) Arbitrariness:

No inherent connection between words and their meanings (e.g., the word "dog" doesn't resemble a dog).

(ii) Productivity:

We can create and understand infinite new sentences (e.g., "The purple cat danced on the moon").

(iii) Displacement:

We can talk about things not present (e.g., the past, the future, imaginary things).

(iv) Duality of Patterning:

Meaningful words are made from meaningless sounds (phonemes → morphemes).

(v) Cultural Transmission:

Language is learned from others, not biologically inherited like blinking.

(vi) Recursion:

Sentences can be nested within others: “I know that you know that she left.”

Component	Description
Phonetics/Phonology	Sounds of speech and their patterns
Morphology	Structure of words (prefixes, roots, suffixes)
Syntax	Sentence structure and grammar rules
Semantics	Meaning of words and sentences
Pragmatics	Contextual meaning and language use in conversation

Human Voice Processing in NLP

- Processing human voice in NLP involves converting spoken language into text and then analyzing that text using various NLP techniques.
- This is known as speech-based NLP and happens in multiple steps.

Steps to Process Human Voice in NLP

Step	Name	Description	Example
1	Speech Recognition (ASR - Automatic Speech Recognition)	Converts spoken voice into text	"What is the weather today?" → text: "What is the weather today?"
2	Natural Language Understanding (NLU)	Understands the meaning, intent, entities, and context	Intent: GetWeather Entity: today
3	Response Generation	Generates a reply (textual or spoken) based on understanding	"The weather today is 32°C and sunny."
4	Text-to-Speech (TTS) (optional)	Converts the text response back into spoken voice	Synthesized speech says: "The weather today is thirty-two degrees and sunny."

Technologies and Models Used for Human Voice Processing in NLP

(a) Speech-to-Text (ASR)

Converts voice to text using **acoustic models** and **language models**.

Common tools:

- ✓ Google Speech-to-Text API
- ✓ IBM Watson Speech to Text
- ✓ Mozilla DeepSpeech (open-source)

(b) NLP/NLU Processing

Processes the transcribed text using:

- ✓ **Tokenization**: Splits the sentence into words.
- ✓ **Part-of-Speech (PoS) :** Identifying grammar components (noun, verb, etc.)
- ✓ **Tagging Named Entity Recognition**: Finds names, places, dates.
- ✓ **Intent Detection**: Classifies the user's purpose.

Common Tools:

- ✓ Google Dialogflow
- ✓ Microsoft LUIS
- ✓ Rasa NLU (open source).

(c) Text-to-Speech (TTS)

Converts the generated reply into voice using:

- ✓ WaveNet (by Google DeepMind)
- ✓ Amazon Polly
- ✓ Google Cloud TTS
- ✓ Festival (open-source)

(c) Text-to-Speech (TTS)

Converts the generated reply into voice using:

- ✓ WaveNet (by Google DeepMind)
- ✓ Amazon Polly
- ✓ Google Cloud TTS
- ✓ Festival (open-source)

Challenges in Voice NLP

- ✓ Accents and dialects
- ✓ Background noise
- ✓ Ambiguity in spoken phrases
- ✓ Multilingual or code-mixed speech (e.g., Hindi-English).

Example in a Real-Life System (e.g., Google Assistant)

User says: *"Set an alarm for 6 AM."*

1. **Speech Recognition**: Audio → "Set an alarm for 6 AM"
2. **NLU**:
   - ✓ Intent: SetAlarm
   - ✓ Entity: Time = 6 AM
3. **System Action**: Schedules alarm
4. **Response Generation**: "Your alarm is set for 6 AM."
5. **Text-to-Speech**: Voice says: "Your alarm is set for 6 AM."

1.2 NLP Models

- NLP models are AI algorithms designed to understand, interpret, and generate human language.
- NLP models are trained on vast amounts of text data to perform various language-related tasks.

Popular NLP Model Architectures

Model	Developer	Key Features
BERT	Google	Bidirectional, good for context-based tasks
GPT-3/4	OpenAI	Text generation, strong zero-shot ability
RoBERTa	Facebook AI	Robust version of BERT
T5	Google	Text-to-text framework
XLNet	Google/CMU	Permutation-based, captures context well

Types of NLP Models

Model Type	Description	Examples	Use Cases
Rule-Based Models	Use manually created rules for text analysis.	Regex, Grammar Parsers	Text parsing, tokenization
Statistical Models	Learn from language patterns in large datasets.	HMM, Naive Bayes	POS tagging, sentiment analysis
Machine Learning Models	Use features and labeled data for learning patterns.	SVM, Decision Trees	Spam detection, document classification
Deep Learning Models	Use neural networks for advanced language understanding.	RNN, LSTM, CNN	Translation, speech recognition
Transformer-Based Models	Use attention mechanisms for better context handling.	BERT, GPT, RoBERTa, T5	Text generation, Q&A, summarization
Multimodal Models	Combine text with other inputs like images or audio.	CLIP, Flamingo	Image captioning, video understanding

Common Tasks Solved by NLP Models

Task	Description	Example Tools/Models
Text Classification	Categorize text into predefined classes	SVM, BERT
Named Entity Recognition (NER)	Identify entities like names, places	spaCy, Flair, BERT
Machine Translation	Translate text between languages	Google Translate, mBART
Text Summarization	Produce shorter versions of documents	T5, BART
Question Answering	Answer questions based on context	GPT, BERT
Text Generation	Generate human-like text	GPT-3, GPT-4, LLaMA
Sentiment Analysis	Determine sentiment of a text	TextBlob, RoBERTa
Speech Recognition	Convert speech into text	Whisper, Wav2Vec

1.3 NLP Ambiguity

- Ambiguity in NLP refers to situations where a sentence, word, or phrase can have **multiple interpretations**.
- It's a core challenge in designing NLP systems because humans can often resolve ambiguity using context, but machines struggle without explicit cues.
- It can **mislead models**, especially in tasks like translation, summarization, or question answering.
- Leads to **wrong outputs** unless properly resolved.
- Often requires **world knowledge** or **context awareness** to disambiguate.

(i) Types of Ambiguity in NLP

(a) Lexical Ambiguity

A word has **more than one meaning**.

Example: "He went to the **bank**."

Bank = financial institution

Bank = side of a river

(b) Syntactic Ambiguity (Structural)

A sentence has **multiple possible grammatical structures**.

Example:

"I saw the man with the telescope."

Did I use a telescope to see the man?

Or does the man have the telescope?

(c) Semantic Ambiguity

The **meaning of the sentence is unclear**, even if the structure is clear.

Example: "Visiting relatives can be annoying."

Are the relatives visiting you?

Or are you visiting the relatives

(d) Pragmatic Ambiguity

Meaning depends on **context, intent, or social norms**.

Example: "Can you pass the salt?"

It's a question, but really a polite **request**, not a query about ability.

(e) Anaphoric Ambiguity

Uncertainty about **what a pronoun or noun phrase refers to**.

Example: "John told Mike that **he** won."

Who won? John or Mike?

(f) Attachment Ambiguity

Related to syntactic ambiguity—**unclear where a phrase attaches** in the sentence.

Example: "She watched the movie on the couch."

Is the movie about a couch, or was she sitting on the couch?

How NLP Tries to Handle Ambiguity?

- ✓ **Contextual embeddings** (e.g., from BERT, GPT)
- ✓ **Disambiguation algorithms** (Word Sense Disambiguation)
- ✓ **Parsing techniques** (to determine sentence structure)
- ✓ **Coreference resolution** (for pronoun ambiguity)
- ✓ **Pretraining on large corpora** (so models learn common usage patterns)

Paradigm	Learning	Key Technologies	Typical Era
Rule-Based	None (manual rules)	Symbolic grammars	1950s–1980s
Statistical	Probabilistic	HMMs, N-grams	1980s–2000s
Machine Learning	Supervised	SVM, CRF, NB	2000s
Deep Learning	Representation Learning	RNNs, CNNs	2012–2018
Transformer-based	Pretraining + Fine-tuning	BERT, GPT, T5	2018–now
Instruction-tuned & Multimodal	Human feedback + Cross-modal input	GPT-4o, Gemini, Sora	2023–now

1.4 Phases in Natural Language Processing

- Natural Language Processing (NLP) is a task of processing human language is typically broken down into phases or stages, each of which handles a specific aspect of language understanding or generation.

- The main phases in NLP:
  - ✓ Text Preprocessing
  - ✓ Lexical Analysis
  - ✓ Syntactic Analysis (Parsing)
  - ✓ Semantic Analysis
  - ✓ Discourse Analysis
  - ✓ Pragmatic Analysis.

Phase	Purpose	Key Tasks
Text Preprocessing	Clean and prepare raw text	Tokenization, stopword removal
Lexical Analysis	Analyze individual words/tokens	POS tagging, NER
Syntactic Analysis	Identify grammatical structure	Parsing, chunking
Semantic Analysis	Extract meaning	WSD, semantic roles
Discourse Analysis	Understand text across sentences	Coreference, cohesion
Pragmatic Analysis	Interpret based on context	Intent detection, irony

1.5 NLP Processing Paradigms

- NLP models are AI algorithms designed to understand, interpret, and generate human language.
- NLP models are trained on vast amounts of text data to perform various language-related tasks.
- Different NLP approaches or strategies used to analyze and interpret natural language.
- NLP paradigms have evolved over time from rule-based systems to modern AI-driven methods.
- NLP process refers to a step-by-step pipeline that transforms raw human language into structured, machine-understandable data.
- Each step helps refine and extract meaning from the text.

Evolution of Paradigms

1950s–1980s: Rule-based → symbolic logic, grammars

1990s–2000s: Statistical NLP → use of large corpora, probabilistic models

2010s: Machine learning → supervised/unsupervised models for various NLP tasks

2015–Present: Deep learning & transformers → contextual, scalable, pre-trained models

Use Case	Best Paradigm(s)
Grammar checking	Rule-based / Hybrid
Spam detection	ML / Statistical
Sentiment analysis	Deep learning / Transformer
Chatbots & assistants	Transformer / Hybrid
Language translation	Transformer

NLP Process Flow

Step	Description	Example Tools
1. Text Acquisition	Collecting raw text from sources (web, emails, PDFs, etc.)	Web scraping, OCR, APIs
2. Text Cleaning	Removing noise like punctuation, HTML tags, emojis	Regex, BeautifulSoup, re module
3. Tokenization	Splitting text into words, phrases, or sentences	word_tokenize (NLTK), spaCy
4. Normalization	Standardizing text – lowercasing, stemming, lemmatization	Porter Stemmer, WordNet Lemmatizer
5. Stop Word Removal	Removing common, non-informative words	NLTK, spaCy
6. POS Tagging	Identifying parts of speech (noun, verb, etc.)	NLTK, spaCy
7. Named Entity Recognition (NER)	Identifying names, places, organizations, dates, etc.	spaCy, Stanford NER
8. Syntactic Parsing	Analyzing grammatical structure (dependency parsing)	spaCy, CoreNLP
9. Semantic Analysis	Understanding the meaning (context, sentiment, etc.)	BERT, TextBlob, GPT
10. Task-Specific Output	Final stage (e.g., classification, translation, summarization)	BERT, GPT, T5

NLP Processing Paradigms

Paradigm	Description	Key Features	Examples
Rule-Based	Uses handcrafted rules and linguistic knowledge	Deterministic, interpretable	Regex, context-free grammar
Statistical	Learns patterns from large corpora using probability and statistics	Data-driven, needs labeled data	HMM, Naive Bayes, n-gram models
Machine Learning	Uses supervised/ unsupervised ML to classify or predict text-related patterns	Feature engineering required	SVM, Decision Trees, CRF
Deep Learning	Uses neural networks to learn from raw text	Minimal feature engineering, high performance	RNN, LSTM, CNN
Transformer -Based	Uses attention mechanisms to capture long-range dependencies	Parallelizable, context-aware	BERT, GPT, RoBERTa, T5
Hybrid	Combines rule-based, statistical, and neural approaches	Balanced performance and interpretability	spaCy (rules + ML), Watson NLU
Multimodal NLP	Processes text along with images, audio, or video	Cross-modal learning	CLIP, Flamingo, GPT-4o

1.6 Need for NLP and NLU

- NLP (Natural Language Processing) is the Toolbox
- NLU (Natural Language Understanding) is the understanding layer that uses tools.

Need NLP and NLU to:

- ✓ Enable machines to communicate with humans
- ✓ Extract meaningful insights from text
- ✓ Power smart, language-based systems
- ✓ Handle the complexity of real-world language use.

(i) Bridging the Gap Between Humans and Machines

- **Human language is messy** — full of ambiguity, slang, idioms, and unstructured data.
- Computers don't "understand" language natively; they need NLP to make sense of it.
- **NLP** helps convert natural language into structured data.
- **NLU** takes it further by trying to understand **meaning and intent**.

Example:

When you say *"Book me a table at 7"*, NLU helps the system realize this is a restaurant reservation, not a request to read a book.

(ii) Information Extraction From Massive Text Data

- Over 80% of digital data is unstructured, mostly in the form of text (emails, documents, social media, etc.).
- NLP lets systems extract useful data: names, locations, topics, sentiments, trends.
- NLU helps identify what is being said and why, even with nuance.

Example:

From a product review, NLP extracts the product name; NLU detects if the user is satisfied or angry.

(iii) Building Intelligent Applications

- NLP and NLU power the core of many everyday tools:

Application Area	How NLP/NLU Helps
Chatbots & Voice Assistants	Understand user questions, commands
Search Engines	Interpret queries and rank relevant results
Translation	Convert meaning from one language to another
Healthcare	Analyze doctor notes, medical reports
Finance	Process legal/financial documents
Customer Service	Analyze support tickets, route requests

(iv) Handling Ambiguity and Context

- Language is contextual and ambiguous.
- Words can have many meanings.
- NLP + NLU are needed to resolve ambiguity, track references, and understand intent.

Example:

"Bank" = riverbank or financial institution?

"He said he would call me back" — who is "he"?

NLU solves this.

(v) Enabling Multilingual and Inclusive Technology

- NLP allows global apps to interact with users in many languages.
- NLU helps understand slang, dialects, or low-resource languages by learning context.

1.7 Relevance of NLP and NLU

Difference Between NLP and NLU

Feature	NLP	NLU
Scope	Handles language processing	Focuses on understanding
Goal	Analyze structure of language	Extract meaning and intent
Includes	Tokenization, POS tagging, parsing	Sentiment, intent, entity linking
Example task	Breaking a sentence into parts	Determining if it's a complaint

Area	NLP/NLU Relevance
Business & Marketing	Customer sentiment, chatbots, automation
Healthcare	Extract data from clinical notes, understand symptoms
Finance & Law	Process contracts, detect fraud, extract insights
Education	Intelligent tutoring systems, grammar feedback
Accessibility	Speech recognition, translation, assistive tech
Social Media	Moderation, trend analysis, emotion detection

1.8 Applications of NLP

- NLP is a part of AI that aims to teach the human language with all its complexities to computers.
- Machines can understand and interpret the human language to eventually understand human communication in a better way.
- NLP is a cross among many different fields such as artificial Intelligence, computational linguistics, human-computer interaction, etc.
- NLP allows computers to understand human language and then analyze huge amounts of language-based data in an unbiased way.
- As digital transformation continues to drive the need to process unstructured data across every industry, artificial intelligence solutions incorporating NLP are becoming business-critical for their ability to improve efficiency, reduce risk and costs and help companies create new opportunities.

Applications:

- ✓ Chatbots
- ✓ Language Translation
- ✓ Search Results / Autocomplete in Search Engines
- ✓ Voice Assistants / Smart Assistants
- ✓ Sentiment Analysis
- ✓ Text Analytics / Text Summarization
- ✓ Grammar Checkers / Plagiarism Checkers
- ✓ Email Classification and Filtering.

(a) Chatbots

- Chatbots simulate human conversations using NLP, understanding both text and sometimes speech input.
- NLP enables chatbots to understand and respond to human language, used in customer support, websites, and virtual agents.
- Chatbots use NLP to interpret user queries and respond in natural language.

NLP Technique Used:

- ✓ **Intent recognition**: Detects what the user wants (e.g., "book ticket" vs. "cancel ticket").
- ✓ **Named Entity Recognition (NER)**: Extracts names, dates, locations (e.g., “book a flight to **Delhi** on **Monday**”).
- ✓ **Dialogue management**: Maintains context across multiple turns in conversation.

Examples:

- ✓ **Banking**: SBI's or HDFC's virtual assistant answers balance inquiries.
- ✓ **E-commerce**: Amazon’s chatbot helps with order tracking or returns.
- ✓ **Healthcare**: ChatGPT-powered bots answer basic medical questions.

(b) Language Translation

- Automatically converts text/speech from one language to another.
- Tools like Google Translate use NLP to translate text from one language to another while preserving meaning and grammar.
- NLP helps translate text between languages, maintaining meaning and grammar.

NLP Techniques Used:

- ✓ **Sequence-to-sequence models** (e.g., Transformer models like Google's BERT or OpenAI's GPT).
- ✓ **Contextual word embedding**: Understands context, e.g., “bank” as a financial institution vs. riverbank.

Examples:

- ✓ **Google Translate**: Converts "How are you?" to "¿Cómo estás?" in Spanish.
- ✓ **Microsoft Translator**: Used in Skype to provide real-time translated subtitles.
- ✓ **Facebook Auto-Translation**: Translates comments/posts into the viewer’s language.
- ✓ **Indian Government’s Bhashini initiative**: Language translation for regional Indian languages.

(c) Search Results / Autocomplete in Search Engines

- NLP helps search engines understand user intent and suggest relevant completions or corrections as you type.
- NLP predicts and suggests search queries as you type, and ranks results based on intent.
- Predicts and ranks what you're likely to search, and displays the most relevant results.

NLP Techniques Used:

- ✓ **Autocomplete using n-grams or deep learning**.
- ✓ **Intent detection and semantic search**: Matches the meaning, not just keywords.

Examples:

- ✓ Typing "weather in" into **Google** shows "weather in Chennai," "weather in Delhi" etc. Type "how to" → suggests "how to file ITR in India 2025."
- ✓ **Bing** suggests "best smartphones under ₹20,000" based on typing behavior and trends.
- ✓ **YouTube**: Autocomplete based on popular and personal watch history.
- ✓ **Quora search**: Suggests relevant questions while typing.

(d) Voice Assistants / Smart Assistants

- Understands spoken language and performs tasks or answers questions.
- Siri, Alexa, and Google Assistant use speech recognition and NLP to process voice commands and respond intelligently.
- Voice input is converted to text (speech-to-text), processed via NLP, then responded to in speech.

NLP Techniques Used:

- ✓ **Speech-to-text and text-to-speech**.
- ✓ **Intent recognition + slot filling** (e.g., "Set alarm for 7 AM" → time = 7 AM).

Examples:

- ✓ **Siri (Apple)**: "What's the weather like today?"
- ✓ **Google Assistant**: "Remind me to take medicine at 8 PM."
- ✓ **Alexa (Amazon)**: "Play devotional songs."

(e) Sentiment Analysis

- Identifies the emotional tone (positive, negative, or neutral) in text.
- This analyzes text (e.g., reviews, tweets) to determine if the sentiment is positive, negative, or neutral.
- Determines whether text expresses a positive, negative, or neutral emotion.

NLP Techniques Used:

- ✓ **Text classification models** (e.g., logistic regression, LSTM, transformers).
- ✓ **Emotion tagging**: Anger, joy, sadness, etc.

Examples:

- ✓ **Twitter monitoring tools**: Identify public reaction to a new government policy.
- ✓ **Product reviews**: "This phone is terrible" → Negative sentiment.
- ✓ **Customer feedback**: Analyzed by companies to improve services.
- ✓ **Brand monitoring**: Zomato uses it to assess customer sentiment on reviews.
- ✓ **Stock market**: Traders use news sentiment for predictive analysis.

(f) Text Analytics / Text Summarization

- NLP can extract key points from large texts and generate concise summaries, useful in news, legal, and academic fields.
- Extracts insights from large documents or summarizes long texts into shorter versions.
- Finds key information in text and summarizes long content.

NLP Techniques Used:

- ✓ **Extractive summarization**: Pulls key sentences (e.g., TextRank algorithm).
- ✓ **Abstractive summarization**: Rewrites the summary using its own words (e.g., GPT-based models).

Examples:

- ✓ **News summarizers**: Provide a 3-line summary of a full article (e.g., Inshorts app).
- ✓ **Legal document analysis**: Tools like Ross Intelligence summarize lengthy case laws.
- ✓ **Academic research tools**: Like SciSummary summarize scientific papers and Scholarcy summarize research papers.

(g) Grammar Checkers / Plagiarism Checkers

- Corrects grammar, punctuation, style, and detects duplicate content.
- Tools like Grammarly use NLP to detect grammar issues, style problems, and duplicated content.
- Detects grammatical errors, punctuation issues, and checks for copied content.

NLP Techniques Used:

- ✓ **Syntax and parsing**: Detects sentence structure.
- ✓ **Paraphrase detection**: Identifies rewritten content that's still plagiarized.

Examples:

- ✓ **Grammarly**: Suggests changes like "He go to school" → "He goes to school."
- ✓ **QuillBot or Turnitin**: Detects plagiarism in student assignments or online content.

(h) Email Classification and Filtering.

- Sorts emails by category and filters spam/phishing emails.
- Emails are analyzed to classify them (e.g., spam, promotions, social) using NLP and machine learning.
- NLP categorizes emails and identifies spam or promotional content.

NLP Techniques Used:

- ✓ **Spam detection** using Bayesian filters or ML classifiers.
- ✓ **Text classification models** to label emails as Social, Promotions, Updates, etc.

Examples:

- ✓ **Gmail**: Separates inbox into Primary, Social, Promotions, and Spam.
- ✓ **Outlook**: Flags phishing emails based on suspicious text patterns.
- ✓ **Email clients**: Auto-label emails with tags like “Work,” “Travel,” etc.

1.9 Understanding of Language

(natural language, symbolic language, computer language)

Natural Language:

- Focus of NLP.
- They are complex, context-rich, and ambiguous, requiring deep learning models (like BERT, GPT) to interpret meaning.

Symbolic Language:

- Often used in **Symbolic AI** (older AI approach),
- Logic and rules are used to simulate intelligence (e.g., expert systems).

Computer Language:

- NLP is sometimes used to understand code comments or translate natural language into code (like in code generation or AI-assisted coding tools).

Aspect	Natural Language	Symbolic Language	Computer Language
Definition	Human spoken or written languages	Abstract representation using symbols	Languages used to program computers
Examples	English, Hindi, Tamil, Telugu, Sanskrit, Punjabi, Marathi, French, etc.	Mathematical logic, chemical equations, music notation	Python, Java, C++, HTML
Users	Humans	Humans (especially in scientific fields)	Programmers, developers
Purpose	Communication among people	Precise symbolic reasoning or expression	Instructing machines/computers
Ambiguity	High (depends on context)	Medium (requires interpretation)	Very low (strict syntax and semantics)
Interpretation	Requires context and world knowledge	Requires domain-specific rules	Requires compilers or interpreters
Processed by	NLP systems, humans	Logic solvers, symbolic AI, humans	Compilers, interpreters
Flexibility	Very flexible	Moderately flexible	Rigid structure and syntax
Structure	Unstructured or semi-structured	Structured	Fully structured

1.10 Expert System in AI

- An Expert System in Artificial Intelligence (AI) is a computer program designed to simulate the decision-making ability of a human expert.
- Built to solve complex problems by reasoning through bodies of knowledge, represented mainly as if-then rules.
- Expert systems are widely used across industries to simulate human expertise in decision-making.

Work Flow:

- **User Input**: User provides a query or problem description.
- **Inference Engine**: Matches input with rules in the Knowledge Base.
- **Reasoning**: Uses forward chaining or backward chaining to reach a conclusion.
- **Output**: Presents solution/recommendation, optionally with an explanation.

Components of an Expert System

Component	Description
Knowledge Base	Contains facts and rules about the specific domain (e.g., medical, legal).
Inference Engine	Applies logical rules to the knowledge base to deduce new information or make decisions.
User Interface	Allows users to interact with the system, inputting queries and receiving advice.
Explanation Facility	Provides reasoning behind conclusions (helps users trust the system).
Knowledge Acquisition	Allows experts to input new knowledge or modify existing rules.

Types of Expert Systems

Type	Key Feature	Example	Application
Rule-Based	IF-THEN rules	MYCIN, XCON	Medical, configuration
Fuzzy	Uncertainty handling	FuzzyCLIPS	Climate control, risk analysis
Frame-Based	Object/attribute structure	Medical frames	Engineering, biological data
Model-Based	System modeling	Automotive diagnosis	Fault detection, simulations
Case-Based	Uses past cases	Legal case retrieval	Legal, help desks
Hybrid (Knowledge-Based)	Combines rules, ML, NLP	IBM Watson	Healthcare, business intelligence

Applications of Expert Systems in AI

Domain	Application Example	Benefit	Tools / Systems Used
Medical	Disease diagnosis, treatment advice	Fast, expert-level medical support	MYCIN, IBM Watson Health, DXplain
Finance	Credit scoring, fraud detection	Risk reduction, better decisions	Expert System (ES) Finance, FinCEN AI, SmartStream
Agriculture	Crop disease detection, soil advisory	Increased yield, reduced loss	AgriExpert, PestDx, Agrobase
Legal	Legal research, case analysis	Time and cost savings	LEXIS, Jurisprudence Expert, CaseMap
Education	Intelligent tutoring, grading systems	Personalized learning	SMILE, SHERLOCK, ETITS
Security	Threat detection, surveillance analysis	Early threat prevention	EMYCIN, CyberES, PROSPECTOR
Environment	Weather prediction, disaster alerts	Safety and sustainability	CLIPS, DENDRAL (chemistry), Flood Expert
Transportation	Traffic management, route planning	Efficiency, reduced travel time	TRANSYT, AutoMechanic Expert, RouteX
Engineering	System configuration, fault diagnosis	Higher reliability, quick fixes	XCON, STES, CAD Expert

1.11 Recommendation System in AI & Cognitive System in AI

(i) Recommendation system/ recommendar system

- A recommendation system is an AI-based algorithm that suggests items to users based on their preferences, behavior, or similarity with other users/items.
- Help the users to find relevant content quickly — like products on Amazon, movies on Netflix, or songs on Spotify.
- Predict what a user might like or need.
- Personalize user experience.
- Increase engagement, sales, or satisfaction.

Benefits:

- ✓ Saves time for users
- ✓ Enhances user satisfaction
- ✓ Filter large volumes of content or products
- ✓ Increases platform retention and revenue
- ✓ Helps users discover new and relevant content and make decision faster.

Types of Recommendation Systems

Type	Description	Example
Content-Based Filtering	Recommends items similar to those the user liked before	Suggests movies with the same genre or director
Collaborative Filtering	Recommends items that similar users liked	"Users who bought this also bought…"
Hybrid Filtering	Combines content-based and collaborative methods	Netflix uses this to recommend shows
Knowledge-Based	Uses explicit knowledge about items and user preferences	Travel or real estate recommendations
Demographic-Based	Uses user attributes like age, gender, location	Suggests music based on age group

Applications:

- ✓ **E-commerce:** Product suggestions (Amazon, Flipkart)
- ✓ **Streaming services:** Movie, TV, music recommendations (Netflix, YouTube, Spotify)
- ✓ **Social media:** Friend suggestions, news feeds (Facebook, Instagram)
- ✓ **News platforms:** Personalized article recommendations
- ✓ **Education:** Course or content recommendations in e-learning platforms

(ii) Cognitive System in AI

- A cognitive system is an AI that simulates human thought processes like learning, reasoning, problem-solving, and understanding natural language.
- Mimic human intelligence for more adaptive, interactive, and contextual computing.

Characteristics:

- ✓ **Learning:** From past data (like Machine Learning)
- ✓ **Reasoning:** Drawing conclusions from data
- ✓ **Natural Language Understanding:** Processing and generating human language
- ✓ **Perception:** Understanding input from sensors or environment (like images, voice)

Examples:

- ✓ **IBM Watson:** Answers complex questions in natural language.
- ✓ **Siri, Alexa, Google Assistant:** Voice-based cognitive systems.
- ✓ **ChatGPT:** Understands and generates human-like responses.

Applications:

- ✓ Healthcare (diagnostic assistants)
- ✓ Finance (risk analysis)
- ✓ Education (AI tutors)
- ✓ Customer service (AI chatbots)

Recommendar System Vs Cognitive System

Feature	Recommendation System	Cognitive System
Purpose	Suggest items to users	Mimic human thought processes
Intelligence Type	Narrow AI (task-specific)	Broad AI (multi-skill, adaptive)
Learning Focus	User preferences and item similarity	Context, reasoning, memory
Example Tool	Netflix's suggestion engine	IBM Watson

1.12 Artificial Intelligence : Introduction

Artificial Intelligence (AI)

- AI is the ability of machines or computer systems to perform tasks that typically require human intelligence.
- These tasks include:
  - ✓ Learning from data
  - ✓ Understanding language
  - ✓ Recognizing images and speech
  - ✓ Solving problems
  - ✓ Making decisions.

Types of AI

- AI can be classified based on capability and functionality.

(i) AI Based on Capability

- It's about how powerful and intelligent the AI is, compared to humans.

(a) Narrow AI (Weak AI)

- AI designed to perform a specific task.
- Most AI today is Narrow AI.
- Does only what it's trained for — cannot generalize.

Examples:

- ✓ Siri / Alexa (voice assistants)
- ✓ Google Translate (language translation)
- ✓ Chatbots, spam filters, facial recognition.

(b) General AI (Strong AI)

- AI with human-level intelligence.
- Can perform any cognitive task that a human can do.
- Learns and applies knowledge across domains.
- Not yet achieved, still under research.

Example (hypothetical):

An AI that can write a novel, solve math problems, cook food, and give legal advice — all like a human.

(c) Super AI

- AI that is smarter than humans in all aspects.
- Can perform better in creativity, emotional intelligence, and reasoning.
- Fully autonomous and possibly self-improving.
- Still theoretical, discussed in future/ethical studies.

Example:

AI that governs countries, creates new technologies, and understands emotions better than humans.

(ii) AI Based on Functionality

- This classification explains **how the AI system works** — what it can do and how it interacts with data and the world.

(a) Reactive Machines

- Only reacts to current input.
- No memory or learning.
- Cannot use past experiences to make decisions.

Example:

- ✓ IBM's Deep Blue (played chess by evaluating positions in real time)

(b) Limited Memory

- Can use past data for a short period.
- Most current AI models fall into this category.

Examples:

- ✓ Self-driving cars (observe recent data like speed, lane, traffic)
- ✓ ChatGPT (remembers previous lines in a conversation).

(c) Theory of Mind (in development)

- Can understand emotions, beliefs, and intentions of others.
- Requires deep understanding of human behavior.
- Still being researched in cognitive AI and robotics.

Example:

A robot therapist who adjusts behavior based on how a person feels.

(d) Self-aware AI (theoretical)

- AI with self-consciousness and awareness of its own existence.
- Can think about its own thoughts and emotions.
- Not yet created.

Example:

A robot that knows it's a robot and makes its own decisions and goals.

Capability Vs Functionality

Type	Description	Example
Narrow AI	Specialized in one task or domain. Cannot perform beyond its scope.	Siri, Google Translate
General AI	Can perform any intellectual task that a human can do. Still theoretical.	Human-level AI (future goal)
Super AI	Surpasses human intelligence in all tasks. Hypothetical.	AGI with emotions, creativity

Type	Description	Example
Reactive Machines	No memory; responds only to current inputs.	IBM's Deep Blue (chess)
Limited Memory	Uses past data for short-term decision making.	Self-driving cars
Theory of Mind	Understands human emotions, beliefs, and intentions.	Still in research
Self-aware AI	Has its own consciousness and self-awareness.	Not yet developed

1.3 Evolution of AI

- The evolution of AI traces the development of machines that simulate human intelligence—reasoning, learning, problem-solving, and decision-making.
- This journey spans from early theoretical ideas to the powerful modern systems like GPT-5 and autonomous vehicles.
- AI evolved from logic-based systems to learning-based models.
- Current AI relies on massive data, neural networks, and compute power.
- AI is now widely applied in real-world domains: healthcare, finance, law, education, transportation, etc.

Timeline of AI Evolution

Era	Milestone	Description
1940s–1950s	Birth of AI (Turing, Logic)	✓ Alan Turing proposed the concept of machines simulating any human activity (Turing Test). ✓ Logic and computation were key foundations.
1956	Dartmouth Conference	✓ Official birth of AI as a field. ✓ Term "Artificial Intelligence" coined by John McCarthy.
1956–1974	Early Enthusiasm	✓ Development of symbolic AI, reasoning systems. ✓ Programs like Logic Theorist and ELIZA emerged.
1974–1980	First AI Winter	✓ Hype failed to match results. ✓ Funding declined due to technical limitations.
1980–1987	Expert Systems Era	✓ Rule-based systems (e.g., MYCIN) used in medicine, engineering. ✓ Commercial success for a short period.
1987–1993	Second AI Winter	✓ Expert systems became brittle and hard to scale. ✓ Funding dropped again.

Contd..

1993–2010	Machine Learning Rise	✓ AI shifted toward data-driven approaches. Algorithms like decision trees, SVMs, and neural nets matured. ✓ IBM's Deep Blue defeated chess champion Garry Kasparov (1997).
2011–Present	AI Boom: Deep Learning & Big Data	✓ Powerful GPUs + Big Data → Deep Learning. ✓ Breakthroughs in NLP (GPT), Vision (CNNs), Speech, Games (AlphaGo).
2020s	Generative AI & AGI Research	✓ GPT-3/4, DALL·E, ChatGPT, Sora. ✓ Focus on explainability, safety, and ethics. ✓ Steps toward Artificial General Intelligence (AGI).

AI Paradigms Over Time

Generation	Paradigm	Examples
1st Gen	Symbolic AI (Logic-based)	Rule engines, Theorem proving
2nd Gen	Knowledge-based Systems	Expert systems, MYCIN
3rd Gen	Machine Learning	Decision trees, SVM
4th Gen	Deep Learning	CNNs, RNNs, Transformers
5th Gen	Generative & Self-supervised AI	GPT-4, DALL·E, ChatGPT, Sora

Modern AI Capabilities

Domain	Example
Language	ChatGPT, Google Translate
Vision	Facial recognition, Medical image diagnosis
Speech	Siri, Alexa, Google Assistant
Robotics	Boston Dynamics, Industrial robots
Games	AlphaGo, OpenAI Five
Generative AI	Text (GPT), Images (DALL·E), Video (Sora)

Text Book(s):

[1] Coppock, E., & Champollion. L. (2019). Invitation to Formal Semantics. Semantics Boot Camp.

[2] Partee, H. B. (1990). Mathematical methods in Linguistics.

[3] Carnie, A. (2002). Syntax: A Generative Introduction. Malden, M A: Blackwell.

[4] Hurford, R. J., Heasley. B., & Smith B. M. (2007). Semantics: A Coursebook. Cambridge University Press, New York.

[5] Jacobson. P. (2014). Compositional semantics:An interoduction to the Syntax and Semantics Interface. Oxford University Press:NY.

[6] Comorobski. I. (1996). Interrogative Pharases: Syntax and Semantics Interface. Sringer:USA.

Thank You !